Riding the Wave?

Who Starts, Follows and Gains from ML Research Trends

A Pre-registered Study of 43,480 ICLR Submissions, 2017–2026

[Your name]

[Department], [Institution]

[Venue / seminar] [Date]

Outline

1. Motivation and Research Questions
2. Data and Measurement
3. Methodological Contribution: Hindsight Bias in Topic Maps
4. Empirical Strategy
5. Results
6. Conclusion

Motivation: Research Fashions in Machine Learning

Share of ICLR submissions that ...	2017	2026
propose a new architecture	24%	8%
build on a foundation model	0.3%	54%
carry a branded title ("NAME: subtitle")	9%	41%

Table 1: Field-wide shifts in content and in style.

- ML research moves in waves, in *what* is studied and in *how* it is presented.
- Are these fashions? If so: what do they displace, who drives them, and are followers rewarded?

Research Questions and Pre-registered Hypotheses

Question	Key hypothesis	Predicted sign
RQ1 Rise: what does a fashion displace?	**H1d**: prior foundation-model (FM) adoption in a topic → share of new-architecture papers	−
RQ2 Following: who introduces new work?	**H2b**: newcomer paper → new-method paper	−
	H2c: newcomer paper → benchmark/evaluation paper	+
RQ3 Reward: are followers rewarded?	**H3a**: topic maturity (prior papers in topic) → acceptance	−

Table 2: Selected hypotheses. All ten registered hypotheses and their verdicts: Appendix A1.

Our RQ2 prediction followed the fashion view: newcomers imitate, insiders innovate.

Related Work and Positioning

Two views of who drives new work

- *Fashion and herding*: ideas spread by imitation (Abrahamson, 1991; Bikhchandani et al., 1992) $\Rightarrow$ newcomers follow
- *Outsider innovation*: new entrants bring new ideas (Azoulay et al., 2019); tradition vs. innovation (Foster et al., 2015)
- Team size: large teams develop, small teams disrupt (Wu et al., 2019)

Meta-research on ML

- Benchmark concentration (Koch et al., 2021)
- Values encoded in ML papers (Birhane et al., 2022)

Measurement challenges we address

1. Survivorship: only published papers observed
2. Hindsight: topics defined on the full period
3. Forking paths (Gelman & Loken, 2014) $\Rightarrow$ pre-registration (Nosek et al., 2018)

Data and Measurement

Construct	Operationalisation	Validation
Corpus	43,480 ICLR submissions, 2017–2026, from OpenReview: accepted, rejected, withdrawn	Held-out split reserved for replication
Topic	SPECTER2 embeddings of title + abstract (Singh et al., 2023); $k = 100$ clusters	Agreement with ICLR areas: NMI 0.30 (keyword baseline 0.27); robust to $k = 75, 150$
Contribution type	LLM with a fixed codebook: method, benchmark/evaluation, empirical, theory, other	Blind second pass: $\kappa = 0.93$
Newcomer paper	No author has an earlier ICLR submission *in the topic* (73% of papers)	Exact OpenReview author IDs
Acceptance	Final decision	Sealed until the single registered run

Table 3: Constructs, measures and checks.

Hindsight Bias in Topic Maps

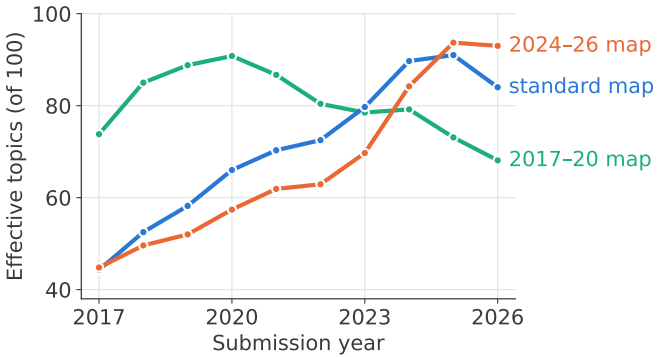


Figure 1: Effective number of topics per year under three $k = 100$ maps fitted on different years.

- 69% of submissions are from 2024–26: a map fitted on all years is a map of the *present*.
- Each map finds peak diversity in the years used to fit it.

Implication

Topic maps are not neutral preprocessing: the fitting window shapes the history one observes.

Consequence: Substantive Conclusions Depend on the Map

- Emerging topics ($\geq 3\times$ growth): **8** with the 2017–20 map vs. **29** with the standard map.
- Do topics age, from new methods towards benchmarks? Same papers, same model:

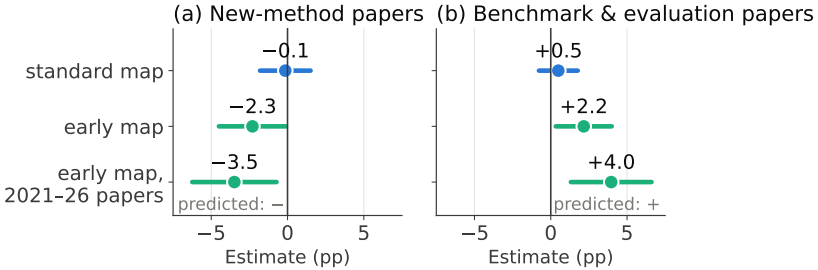


Figure 2: Coefficient on topic maturity (log prior papers), 95% CI. Early-map rows are exploratory.

⇒ Every registered result must also pass two hindsight checks (early-year and year-balanced maps).

Empirical Strategy

Specification (linear probability model)

$$y_i \;=\; \beta\, x_i \;+\; \mu_{k(i)} \;+\; \tau_{t(i)} \;+\; \varepsilon_i$$

- $y_i \in \{0,1\}$: paper type or acceptance; x_i: e.g. newcomer paper
- μ_k, τ_t: topic and year fixed effects (topic $\times$ year for RQ2)
- Standard errors clustered by topic (100 clusters)

Decision rule (registered). Supported only if:

1. predicted sign, and Holm-adjusted $p < 0.05$;
2. same sign in $\geq 75\%$ of 9–11 alternative specifications;
3. same sign under both hindsight checks.

Protocol

Pre-registered; variables built blind to acceptance; pipeline tested on 12,400 simulation runs; code and inputs frozen (SHA-256); a single run; every estimate reported.

Simulation finding

Under realistic no-effect data, clustered SEs rejected up to 12.6% at nominal 5%. The RQ2 results therefore get two extra checks: two-way clustering and a stress test.

RQ1: Foundation-Model Adoption and New Architectures

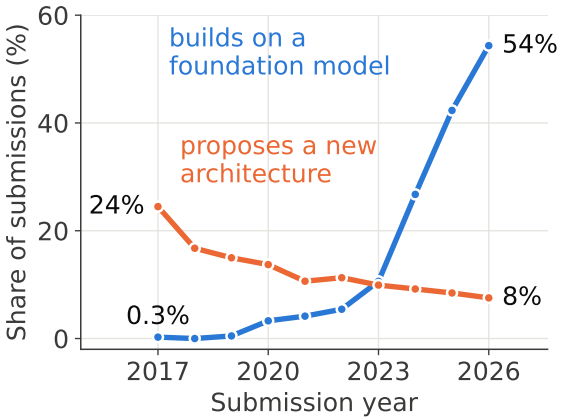


Figure 3: Field-wide trends (descriptive).

Within topic, net of year effects: prior FM adoption vs. new-architecture papers.

$\hat{\beta}$ (0 $\rightarrow$ 100% FM)	-0.080 (0.021)
95% CI	$[-0.121, -0.038]$
Holm p	0.001
Specifications with same sign	9 / 9
N	42,990

- H1d **supported**.
- 10% $\rightarrow$ 50% FM adoption: about one third fewer new-architecture papers (base 9%).

RQ2: Newcomers and Contribution Types

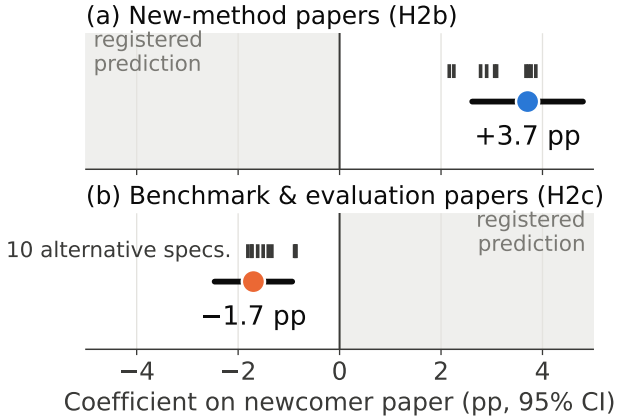


Figure 4: Topic × year fixed effects; grey: registered predicted side.

	H2b (method)	H2c (bench.)
$\hat{\beta}$	$+0.037$	-0.017
(SE)	(0.006)	(0.004)
Base rate	75%	7.7%
Same sign	10/10	10/10

- Both significant opposite to prediction (wild cluster bootstrap $p < 0.001$; $N = 42{,}315$)
- Consistent with outsider innovation, not with herding

Most of the H2c gap goes with team size (-0.005 when controlled); H2b does not.

Exploratory: Fashion in Title Style

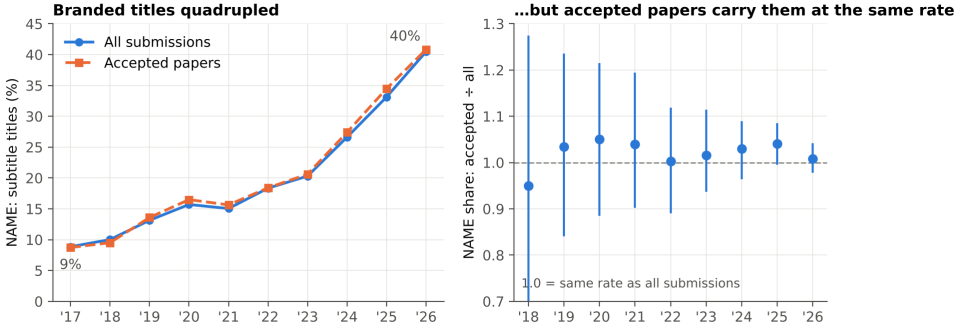


Figure 5: Left: share of branded titles. Right: share among accepted ÷ share among all submissions.

- Branded titles rose more than fourfold; accepted papers carry them at the submitted rate (ratio 1.02).
- The rise appears with and without LLM-style wording in the abstract. (Not pre-registered.)

RQ3: Is Following the Fashion Rewarded?

Association with acceptance	Estimate	Status
Topic maturity, per log unit of prior papers (H3a)	-1.7 pp (SE 1.0)	inconclusive; $\text{MDE}_{80} \approx 4$ pp
Branded title	ratio 1.02	exploratory
New architecture $\times$ prior FM adoption (0 $\rightarrow$ 100%)	$+8.3$ pp (SE 3.5)	exploratory, $p = 0.02$

Table 4: Topic and year fixed effects; SEs clustered by topic; decided papers, 2018–2026 ($N = 32{,}190$ for H3a).

Reading

No evidence that following the fashion is rewarded at review. If anything, papers that go against the tide (architectures where FMs dominate) do relatively better (exploratory).

Other correlates of acceptance (theory papers, team size): Appendix A7–A8.

Which Areas Are Accepted More or Less Often? (Post Hoc)

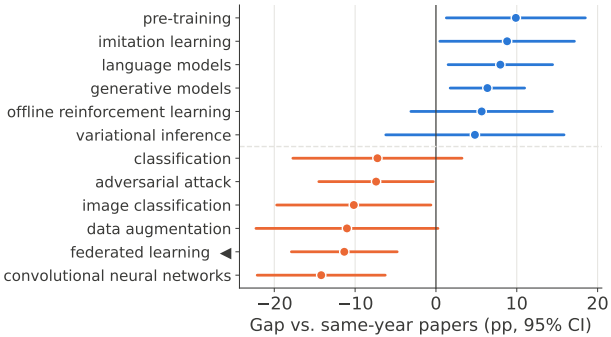


Figure 6: Acceptance gap vs. papers of the same year, 2018–2026. Six highest and six lowest of the author keywords with ≥ 5 years of data.

- Screen of 68 author keywords and 100 topics, 32,190 decided papers
- Areas differ by about ± 10 pp: generative and pre-training work above, federated learning, CNNs and image classification below
- Federated learning is low under *both* definitions: keyword -11 pp, topic -14 pp ($p = 0.005$ each)

No single area survives correction for 68 comparisons ($q \geq 0.17$; conservative test).

Why Is Federated Learning Falling Out of Favour? (Post Hoc)

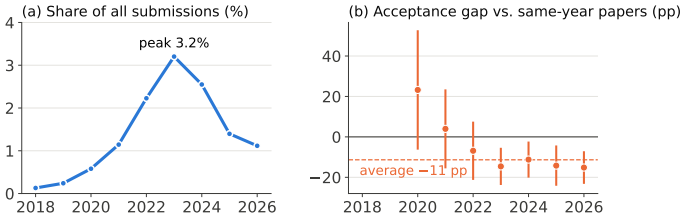


Figure 7: Federated learning (author keyword); 95% CI, years with ≥ 10 papers.

Candidate explanation	Evidence	Verdict
Influx of inexperienced authors	Authors became *more* established (4% → 60%); type and team size stable; adjusting keeps 76% of the decline	No
The field moved on (hype cycle)	Share of submissions peaked in 2023 (3.2%), 1.1% by 2026	Yes
Papers judged more harshly	Gap -11 to -15 pp each year since 2023; decline in every author group (n.s.)	Consistent

Contributions

1. **Methodological.** Topic maps fitted with hindsight manufacture emerging topics and can reverse trend conclusions; early-year maps provide a cheap check.
2. **Design.** A pre-registered, outcome-blind, simulation-checked, single-run protocol for science-of-science studies.
3. **Substantive.**
   - Where foundation models spread, new architectures fade (H1d, supported).
   - Newcomer papers lean towards new methods and away from benchmarks, against the herding prediction (H2b/H2c).
   - Following fashions is not rewarded; areas such as federated learning fall out of favour as the field moves on, not because of who writes the papers (post hoc).

Limitations and Future Work

Limitations

- One venue (ICLR)
- Associations, not causal effects
- LLM labels: consistent ($\kappa = 0.93$), not yet human-validated
- 2021 author-ID change undercounts incumbents (expected bias towards zero)

Future work

- Replication on the held-out ICLR split (five hypotheses registered, incl. federated learning)
- Extension to ACL Anthology / ARR
- Human validation of contribution labels
- Review scores: reviewers vs. final decisions

Thank you

Questions and comments welcome

[email]

A1. All Pre-registered Hypotheses

	Prediction	Estimate (95% CI)	Verdict
H1a	fewer new methods in mature topics	-0.1 pp $(-1.8, +1.5)$	not supported
H1b	more benchmarks in mature topics	$+0.5$ pp $(-0.8, +1.8)$	not supported
H1c	more empirical studies in mature topics	-0.3 pp $(-1.3, +0.7)$	not supported
H1d	**fewer new architectures after FM adoption**	$\mathbf{-8.0}$ pp $(-12.1, -3.8)$	**supported**
H1e	more empirical/evaluation after FM adoption	-0.6 pp $(-3.6, +2.4)$	not supported
H2a	fewer incumbents in hot topics	-0.7 pp $(-2.7, +1.2)$	inconclusive
H2b	**newcomers: fewer new methods**	$\mathbf{+3.7}$ pp $(+2.6, +4.8)$	**significant, opposite**
H2c	**newcomers: more benchmarks**	$\mathbf{-1.7}$ pp $(-2.5, -0.9)$	**significant, opposite**
H3a	lower acceptance in mature topics	-1.7 pp $(-3.7, +0.3)$	inconclusive (low power)
H3b	... more so for method papers	-0.4 pp $(-1.5, +0.8)$	inconclusive

Maturity effects are per log unit of prior papers in the topic (2.7× more). H2d/H3c (buzzwords) were dropped before any estimate because the measure failed its blind validation.

A2. RQ2 Robustness

	New methods	Benchmarks & eval.
Main estimate	$+3.7$ pp	-1.7 pp
Range over 10 alternative specifications (all $p < 0.05$)	$+2.2$ to $+3.9$	-1.8 to -0.9
Two-way clustering (topic $\times$ early-map topic)	$+3.7$	-1.7
Team size controlled	$+3.9$	-0.5 ($p = 0.07$)
Only years with consistent author IDs	$+3.7$	-1.7
Excluding the paper's own earlier versions	$+3.8$	-1.7
Corrected for label errors	$+4.1$	-1.8

Base rates: 75% of papers are new methods, 7.7% benchmark or evaluation; 73% are newcomer papers. Confirmatory p from a wild cluster restricted bootstrap (Cameron et al., 2008): $p < 0.001$ for both. The method gap appears in every era (2019–20: $+5.5$ pp).

A3. Calibration of Standard Errors

- Before seeing any result: 12,400 simulation runs, including no-effect scenarios with realistic topic structure.
- When true topic structure cuts across the 100 topics, the standard test rejected 12.6% (H2b) and 10.0% (H2c) of the time at nominal 5%. The wild bootstrap did not fix this.
- Added before the run:
  - two-way clustering (two topic maps at once);
  - a stress test that deflates each t-statistic by the simulated inflation (1.31 / 1.22).
- RQ2 results: $t = 6.7$ and -4.4. After the stress test, Holm $p = 4 \times 10^{-6}$ and 1×10^{-3}.

Lesson

Check the test on data shaped like yours before trusting its p-values.

A4. Validity of Topics and Labels

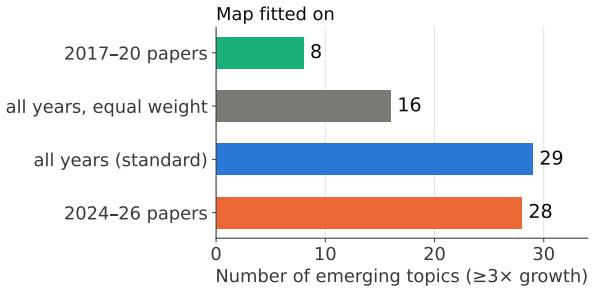
Topics (SPECTER2, 100 clusters)

- Agreement with ICLR's primary areas (2024–26): NMI 0.30 vs. 0.27 for a TF-IDF baseline
- Blind naming test: from title and abstract, a checker picks the right topic 80% of the time at the 100-topic level (61% among 198 fine topics; chance 10%)
- Stable across refits (ARI 0.66)

Contribution labels (LLM, fixed codebook)

- Blind second pass on 2,174 papers: $\kappa = 0.93$
- 95.8% agreement with 2,300 reference labels (also LLM-made, so this shows consistency, not human validity)
- A correction for label errors makes the RQ2 estimates slightly larger

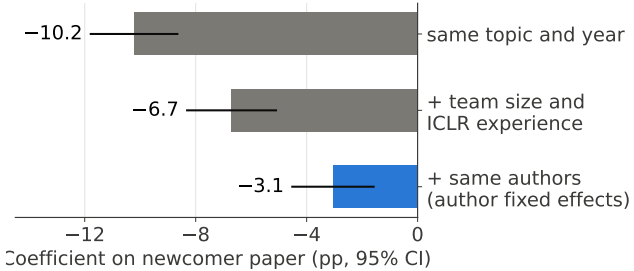
A5. Hindsight: Further Evidence



- Share of 2024–26 papers in emerging topics: 22% with the early map vs. 41% with the standard map
- Declining topics: 13 with the standard map vs. 26 with the year-balanced map, so faded topics are absorbed into today's topics
- Fit of papers to their topic: standard map 0.62 (2017) vs. 0.65 (2026); early map 0.69 vs. 0.53

All maps have 100 topics; only the fitting years differ.

A6. Newcomer Acceptance Gap (Exploratory)

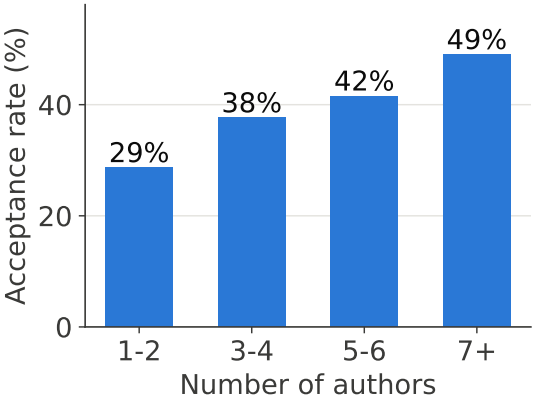


- Raw acceptance: 37% vs. 47%
- The gap shrinks as comparisons tighten; within the same authors it is about -3 pp
- The same person entering a topic new to them: -1.4 pp

Mostly about *who* writes newcomer papers. ICLR is double-blind, so the remainder need not be bias: being new to a topic can make a paper weaker.

Author-FE sample: -5.4 pp without, -3.1 pp with author fixed effects.

A7. Team Size and Acceptance (Exploratory)



- $+8.8$ pp per doubling of team size, within topic $\times$ year
- Steepest for benchmark and evaluation papers ($+13.5$ pp each); this ranking is only suggestive
- Flattest for theory (1–2 authors: 42%)
- Positive in all 20 areas; flattest in federated learning and time series

Team size also proxies compute, data and seniority (Wu et al., 2019).

A8. Acceptance by Contribution Type (Exploratory)

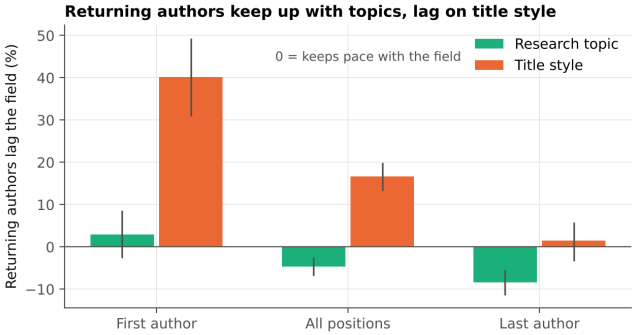
Within topic × year, relative to new-method papers:

Theory	$+8.9$ pp
Dataset / benchmark	$+3.9$ pp
Evaluation	-1.9 pp (n.s.)
Empirical study	-2.7 pp
Other (systems, applications, position papers, . . .)	-8.0 pp

Papers building on a foundation model: $+4.6$ pp vs. papers that do not.

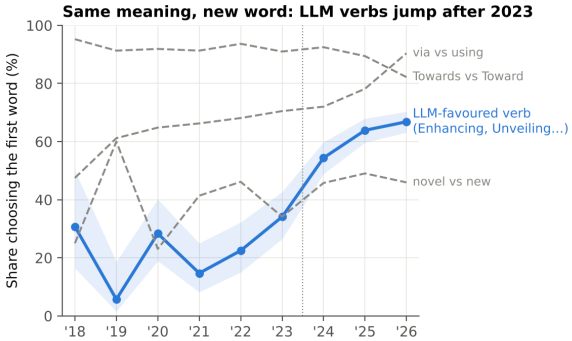
Unadjusted, exploratory. Selection is likely: who writes theory papers differs from who writes method papers.

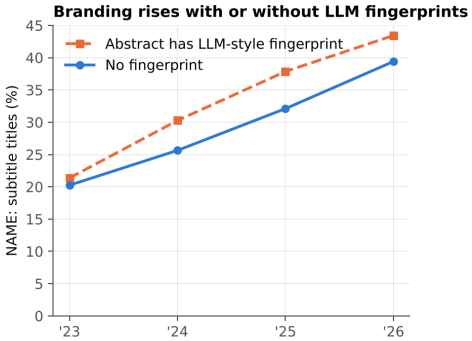
A9. Title Style: Who Carries It? (Exploratory)



- Returning authors move into new topics as fast as the field
- They lag on the new title style, most as first authors
- Topics change with the field; style changes with the people

A10. Title Style and LLM Writing Tools (Exploratory)



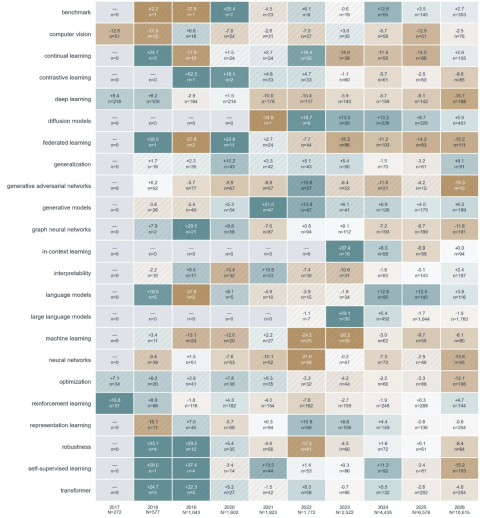


LLM-favoured words ("Enhancing", "Unveiling") jump after 2023 (odds ratio 2.2), while control word pairs do not. The "NAME: subtitle" structure rises with or without LLM fingerprints in the abstract.

A11. Definitions

Topic	one of 100 clusters of paper embeddings (title + abstract)
Topic maturity	$\log(1 +$ earlier papers in the topic$)$; $+1 = 2.7\times$ more papers
Hot topic	top fifth of topics by last year's growth
Newcomer paper	no author has an earlier ICLR submission (any outcome) in that topic
FM adoption	share of the topic's papers last year that built on a foundation model
Benchmark & eval.	primary contribution is a dataset, benchmark or evaluation
Model	linear probability model with topic and year fixed effects (topic $\times$ year cells for RQ2); errors clustered by topic
pp	percentage points

A12. Earlier Keyword × Year Grid (Exploratory, Pre-registration)



Keyword papers minus all others, same year (pp); hatched: a group below 50 papers; 31,239 submissions.

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